Modeling and Analyzing Genotypic Variation in Plant Growth Dynamics

EDEN HUHN, NICHOLAS JOYNER, KATIE THORNTON

Executive Summary

- Model Fitting and Evaluation: The logistic and linear models were most commonly selected as best fits across individual plants.
- Statistical Analysis: The ANOVA revealed significant variation among genotypes, while heritability estimates suggest that linear growth rates are more genetically driven, whereas logistic growth traits are more influenced by environmental factors.
- Clustering Analysis: Clustering revealed that certain genotypes consistently showed faster or larger growth trajectories.

Analysis Procedure

Population Fits
- Fit each growth model across all plants using data averaged by day.
- Store optimized parameters for the population fits for use as initial parameter guesses for individual plants.

Individual Fits
- Loop through individual plants and fit the five growth models to each.
- Store optimized parameters, final cost, and AIC score for each model for each plant.

Calculate Diagnostics
- Calculate R-Squared, Adjusted R-Squared, AIC Weights, and Normalized Probabilities for each model for each plant.

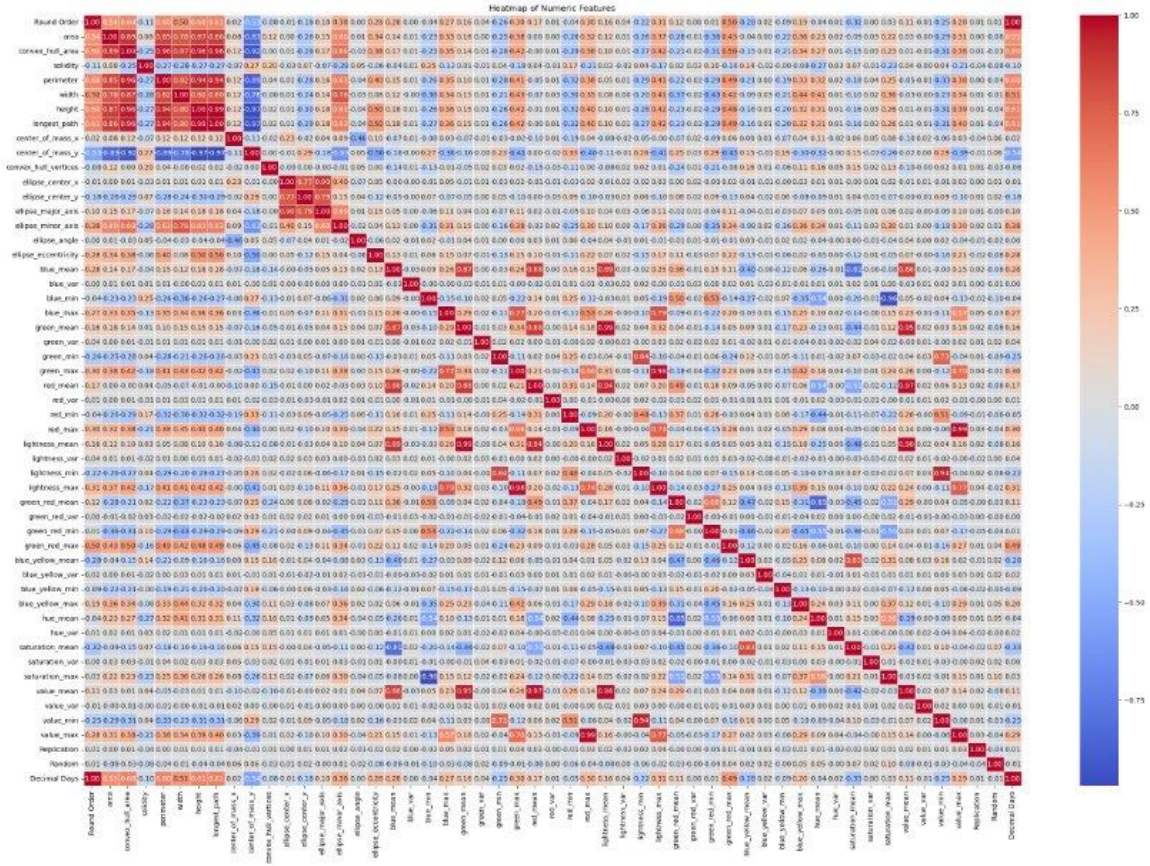
Weighted Growth
- Calculate weighted growth curves by genotype using best fitted models, weighting plants farther from the median less.
- Fit the five growth models to the weighted curves to extract growth parameters.

Statistical Analysis
- Perform ANOVA using optimized logistic and linear growth parameters to compare genotypes.
- Evaluate Heritability of logistic and linear growth parameters.

Clustering Analysis
- Perform K-Means clustering on genotypes using optimized parameters for the weighted growth curves.

Section 1: Exploratory Data Analysis

Correlation (ignoring time series)

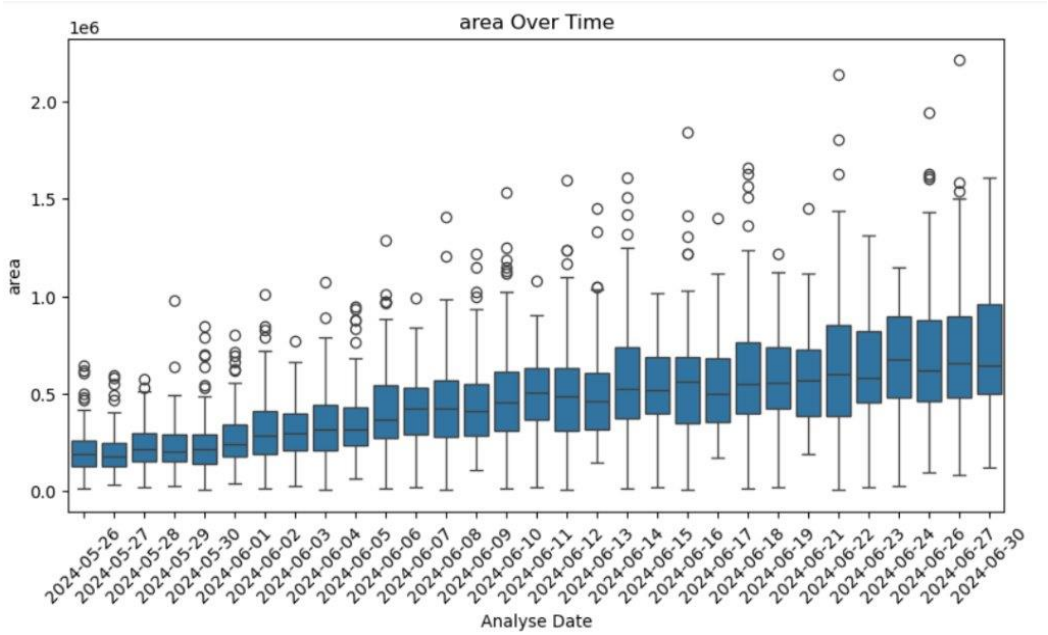


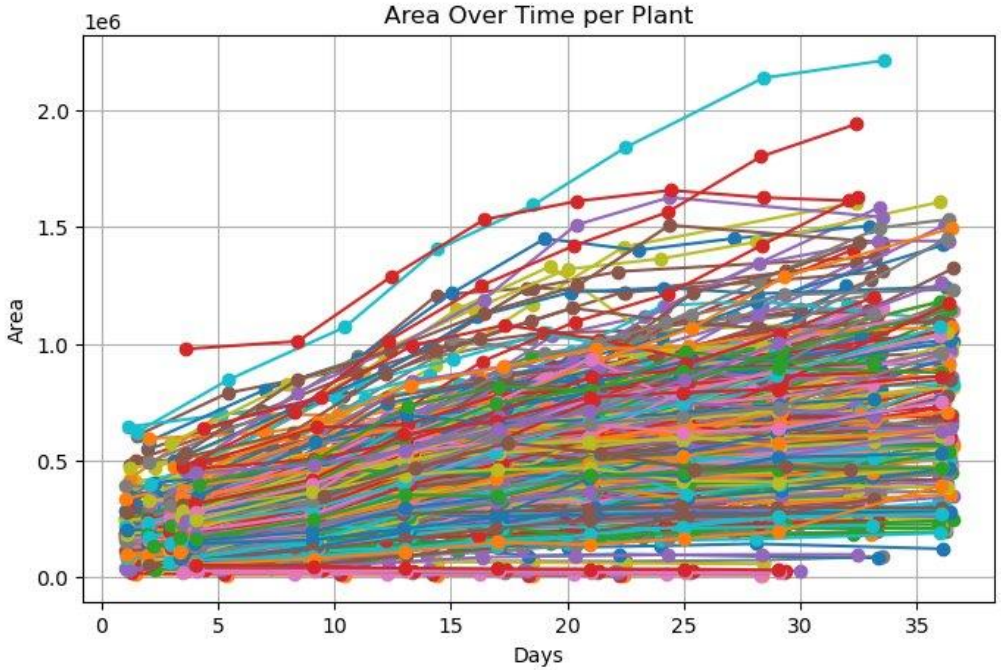
Features	Correlation Coefficient
height, longest path	0.998
green mean, lightness mean	0.995
red max, value max	0.991
green max, lightness max	0.983
lightness mean, value mean	0.980
red mean, value mean	0.971
convex hull area, longest path	0.959
convex hull area, height	0.959
convex hull area, perimeter	0.955
green mean, value mean	0.951

Cross-Correlation for Area (Includes Effect of Time Series)

Feature Associated with Area	Cross-Correlation Coefficient
convex_hull_area	0.854
height	0.802
longest_path	0.796
perimeter	0.784
width	0.714
ellipse_minor_axis	0.503
center_of_mass_y	-0.749

Area Over Time





Biological Takeaways: Variability increases over time, as does plant area. This implies that some trees grow faster or larger than others, at different rates.

Section 2: Model Fitting and Evaluation

Method of Analysis

- Determine models, $f(t)$, appropriate for describing plant growth curves
- Determine optimal parameters, q_{OLS}, for model by minimizing

$$\frac{\sum_{j=1}^{N}\left(y_j - f(t_j; q_{OLS})\right)^2}{N}$$

- Evaluate fit using the Akaike Information Criterion (AIC) and adjusted R2

$$AIC_{OLS} = N \cdot \ln\left(\frac{\sum_{j=1}^{N}\left(y_j - f(t_j; q_{OLS})\right)^2}{N}\right) + 2(k_q + 1)$$ AIC for large data samples

$$AIC_{OLS_c} = AIC_{OLS} + \frac{2(k_q+1)(k_q+2)}{N - k_q}$$ Corrected AIC for small data samples

$$R^2_{Adj} = 1 - (1 - R^2) \cdot \frac{N - 1}{N - (k_q+1)}$$

Method of Analysis

- Calculate AIC weights for growth models – the highest weight represents the best growth model for the data

$$w_i(AIC) = \frac{exp\left(-\frac{1}{2}\Delta_i(AIC)\right)}{\sum_{k=1}^{K} exp\left(-\frac{1}{2}\Delta_k(AIC)\right)}$$

$$\Delta_i(AIC) = AIC_i - AIC_{min}$$

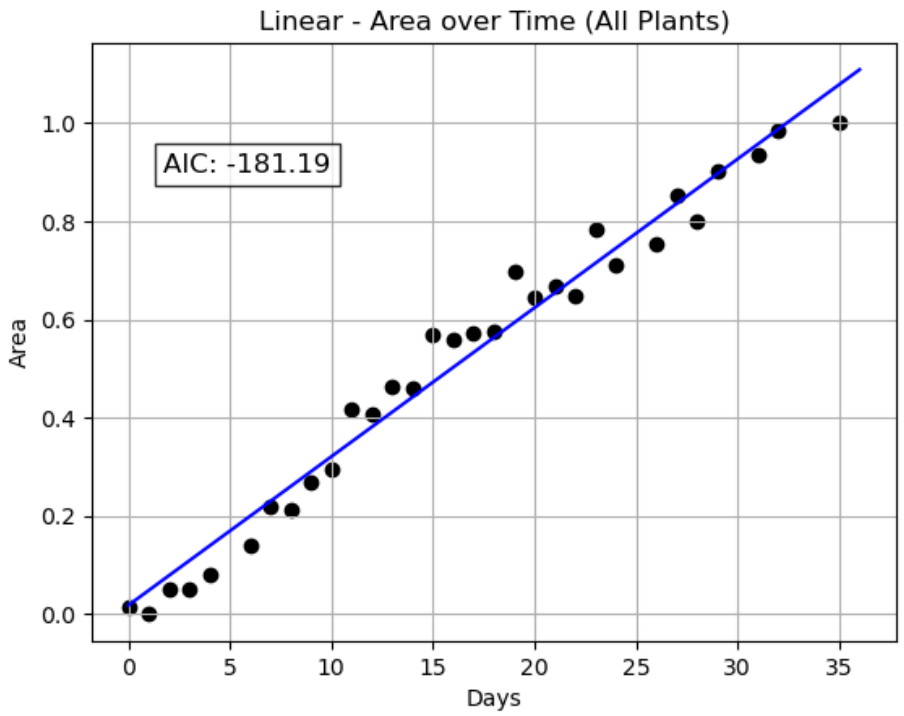
- Use the normalized probability to compare the two best models per plant

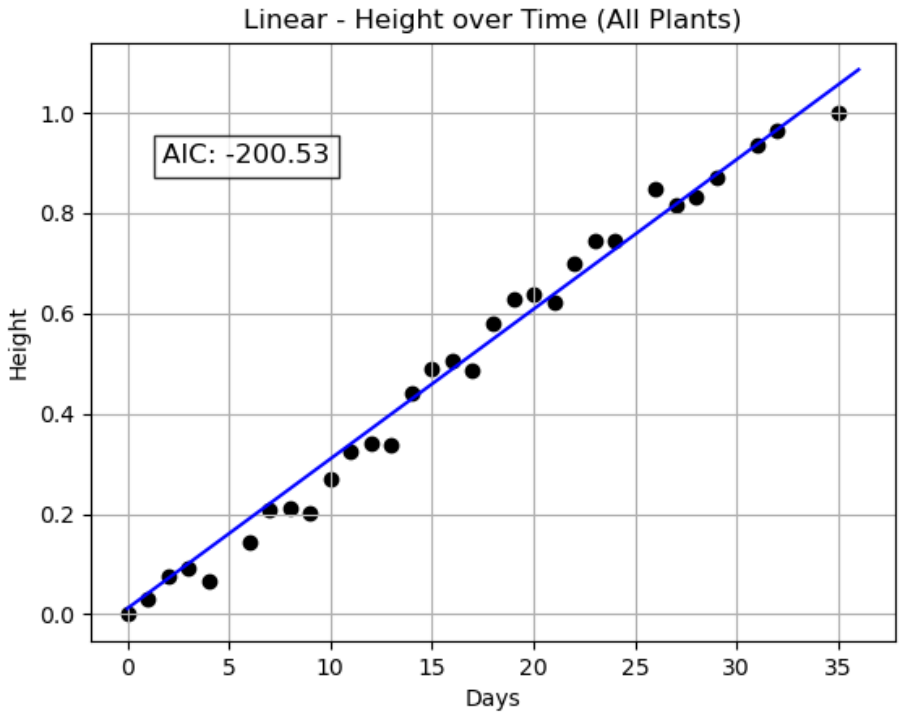
$$Normalized\ Probability = \frac{w_i(AIC)}{w_i(AIC) + w_j(AIC)}$$

w_i(AIC) = AIC weight for the best model

W_j(AIC) = AIC weight for the second-best model

Linear Model

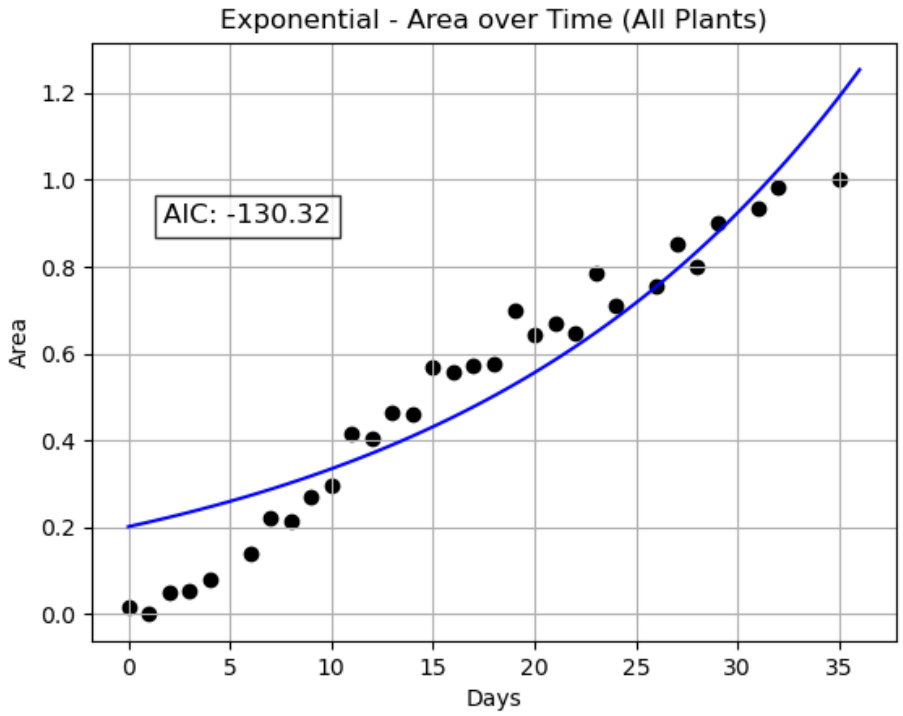




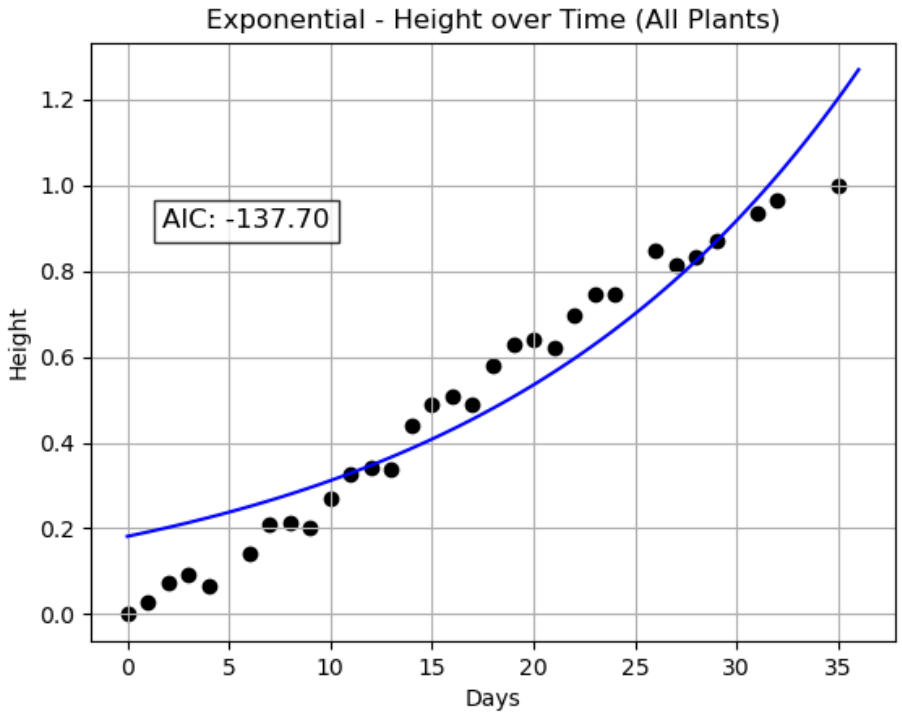
$$y(t) = y_0 + rt$$

- y0 = Initial Population (area/height)
- r = Growth Rate

Exponential Model

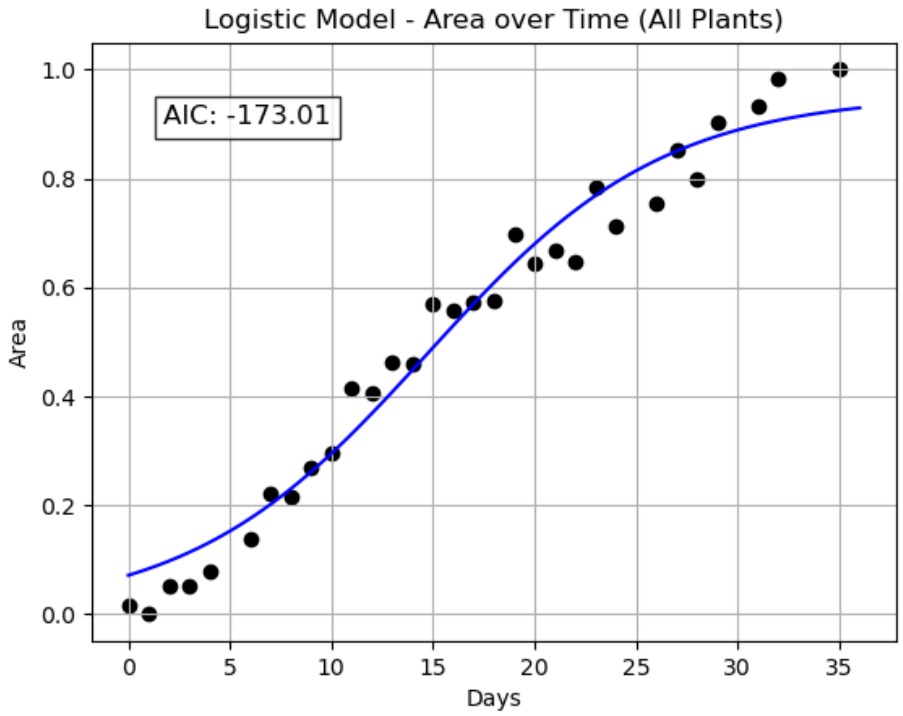


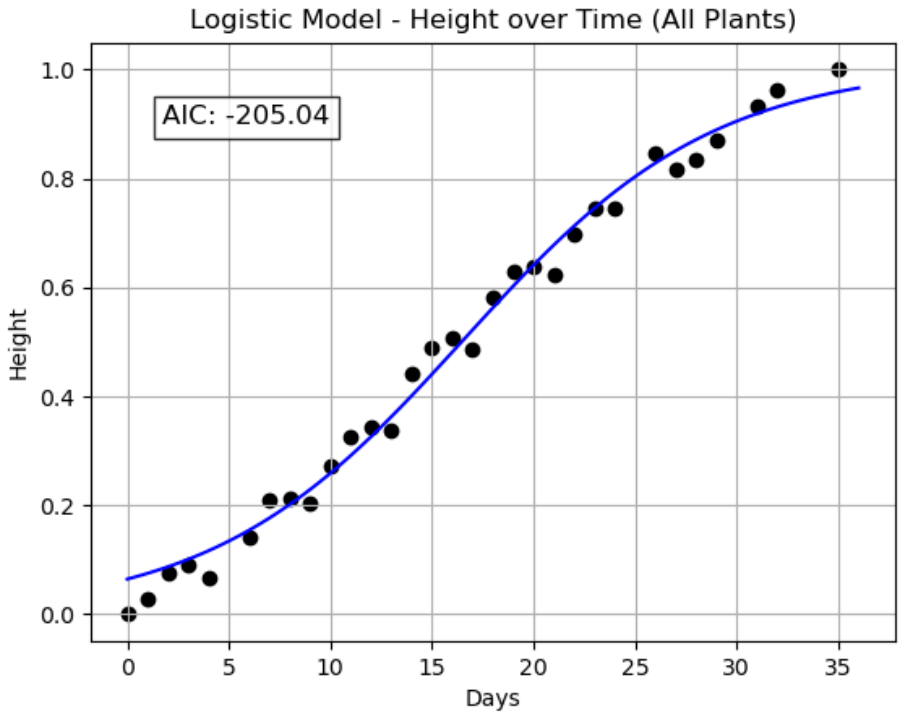
$$y(t) = y_0 \cdot e^{rt}$$



- y_0 = Initial Population (area/height)
- r = Growth Rate

Logistic Growth Model

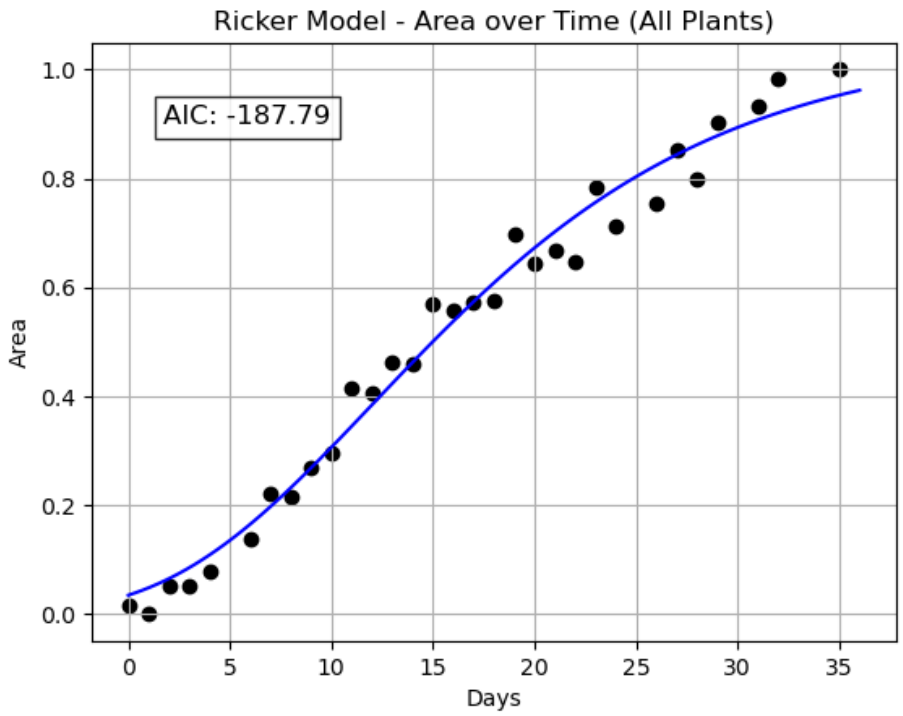


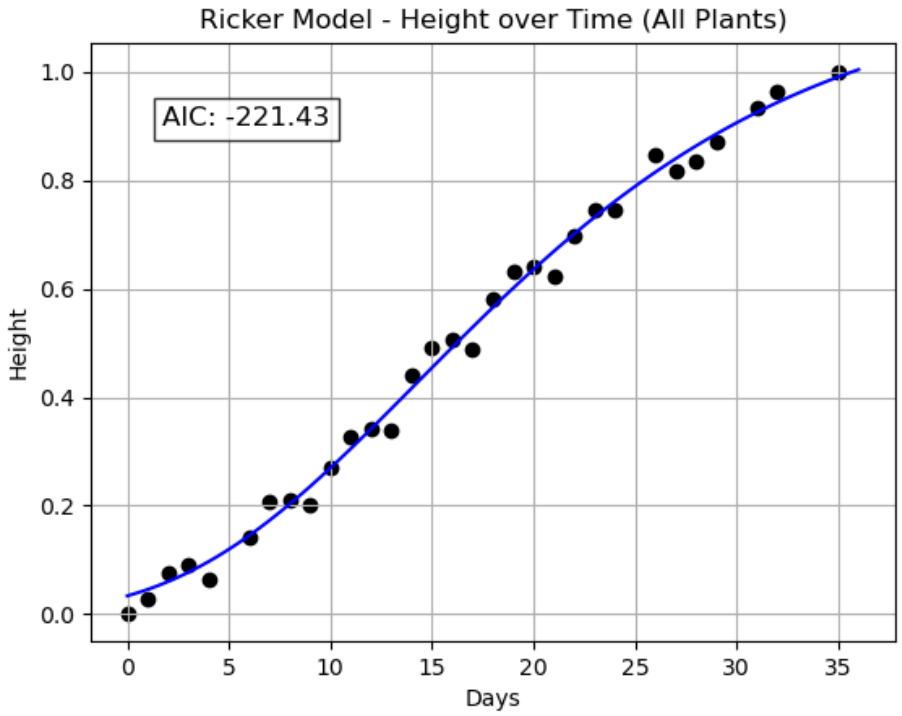


$$y(t) = \frac{Ky_0}{y_0 + (K - y_0)e^{-rt}}$$

- y_0 = Initial Population (area/height)
- K = Limiting Population (area/height)
- r = Growth Rate

Ricker Model

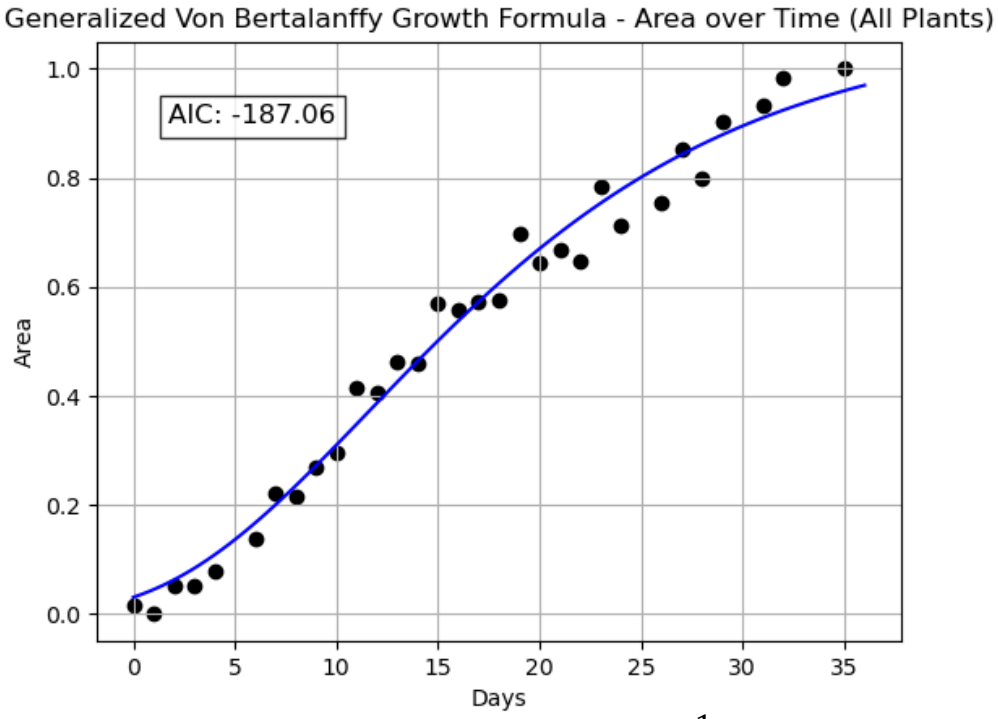


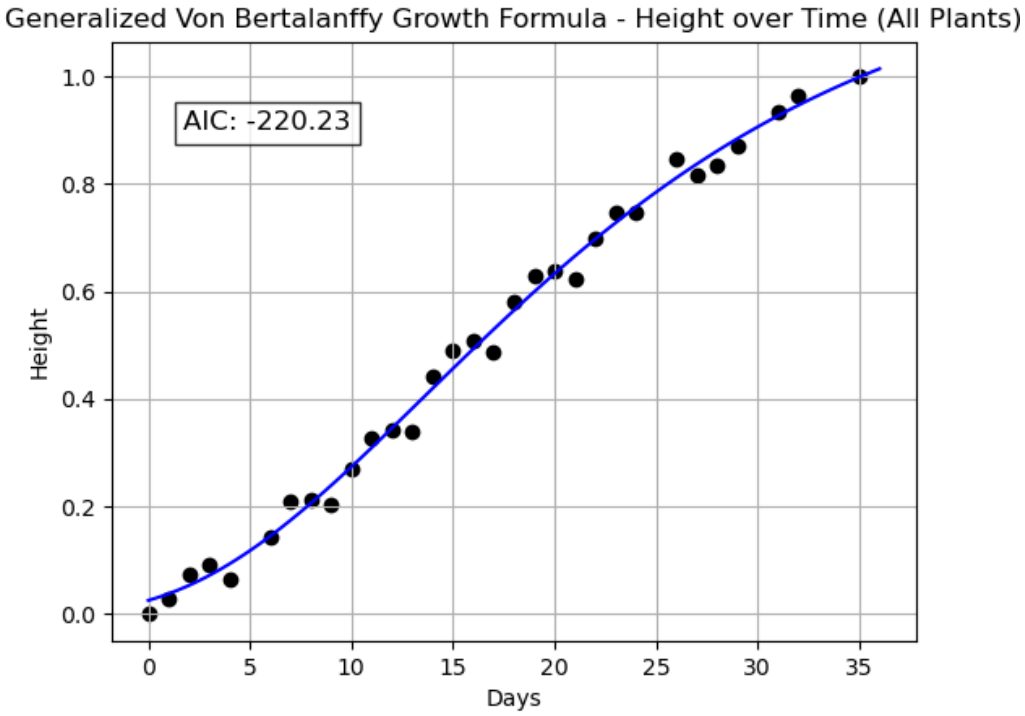


$$y(t) = y_0 \cdot e^{m(1-e^{-rt})}$$

- y_0 = Initial Population (area/height)
- m = Modifies Upper Asymptote
- r = Growth Rate

Generalized Von Bertalanffy Model





$$y(t) = y_0 \left(\frac{1 - f \cdot e^{-rt}}{1 - f} \right)^{\frac{-1}{A-1}}$$

- y_0 = Initial Population (area/height)
- r = Growth Rate
- f = Parameter With No Simple Biological Interpretation
- A = Mass Scaling Exponent

Best Model Frequency for Individual Plants

We determined the best model by calculating the AIC weights for each model on each plant and chose the highest weighted value to assign a best model to each plant.

Area:

Best_Model	
Logistic	174
Linear	167
Exponential	58
Ricker	47

Height:

Best_Model	
Logistic	224
Linear	132
Exponential	74
Ricker	16

Biological Takeaway: The growth of these plants is generally predictable and bounded over time.

Section 3: Statistical Analysis

Methods of Analysis

- Performed analysis across all genotypes using optimal parameters for all plants within genotype.
  - K_{log} = Carrying capacity of logistic curve
  - r_{log} = Rate of logistic growth
  - r_{lin} = rate of linear growth
- One-way analysis of variance (ANOVA) used to determine if there is a statistically significant difference between genotype means for parameter values.

H_0: $\mu_1 = \mu_2 = \ldots = \mu_n$ H_a: At least one genotype mean is different

Methods of Analysis

- Tukey's Honest Significant Difference (HSD) used to perform multiple pairwise comparisons across genotypes

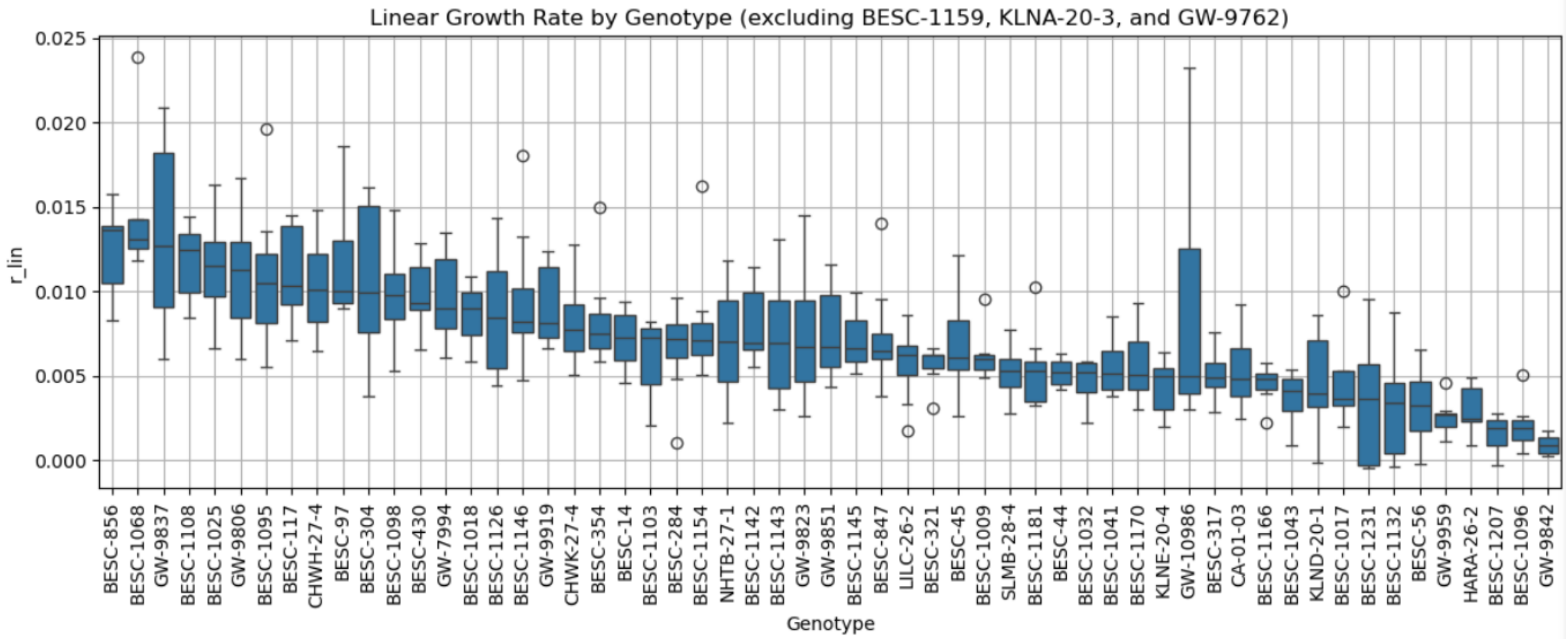
$$HSD = q\sqrt{\frac{MS_W}{N}}$$

- q = value from studentized distribution
- MS_w = mean square within (from ANOVA)
- N = sample size

- Heritability: measures the proportion of total observed variation due to genetic differences

$$\sigma_G^2 = \frac{MS_{Genotype} - MS_{Residual}}{n_r} \qquad \sigma_E^2 = MS_{Residual} \qquad \sigma_P^2 = \sigma_G^2 + \sigma_E^2 \qquad H^2 = \frac{\sigma_G^2}{\sigma_P^2}$$

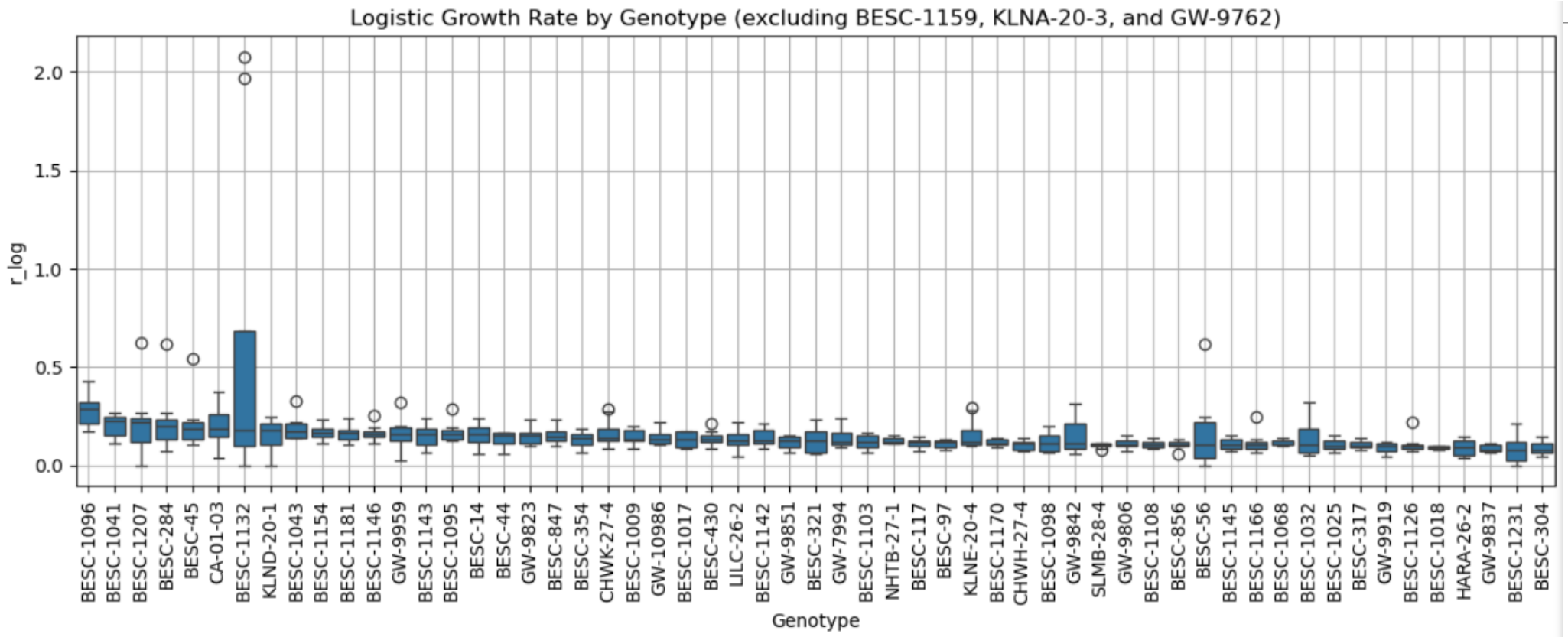
ANOVA results for Linear Growth Rate - Area



Takeaways:

This suggests that growth rate is at least partially influenced by genetic differences.

ANOVA results for Logistic Growth Rate - Area



Note, Tukey HSD revealed several significant pairwise differences for each parameter.

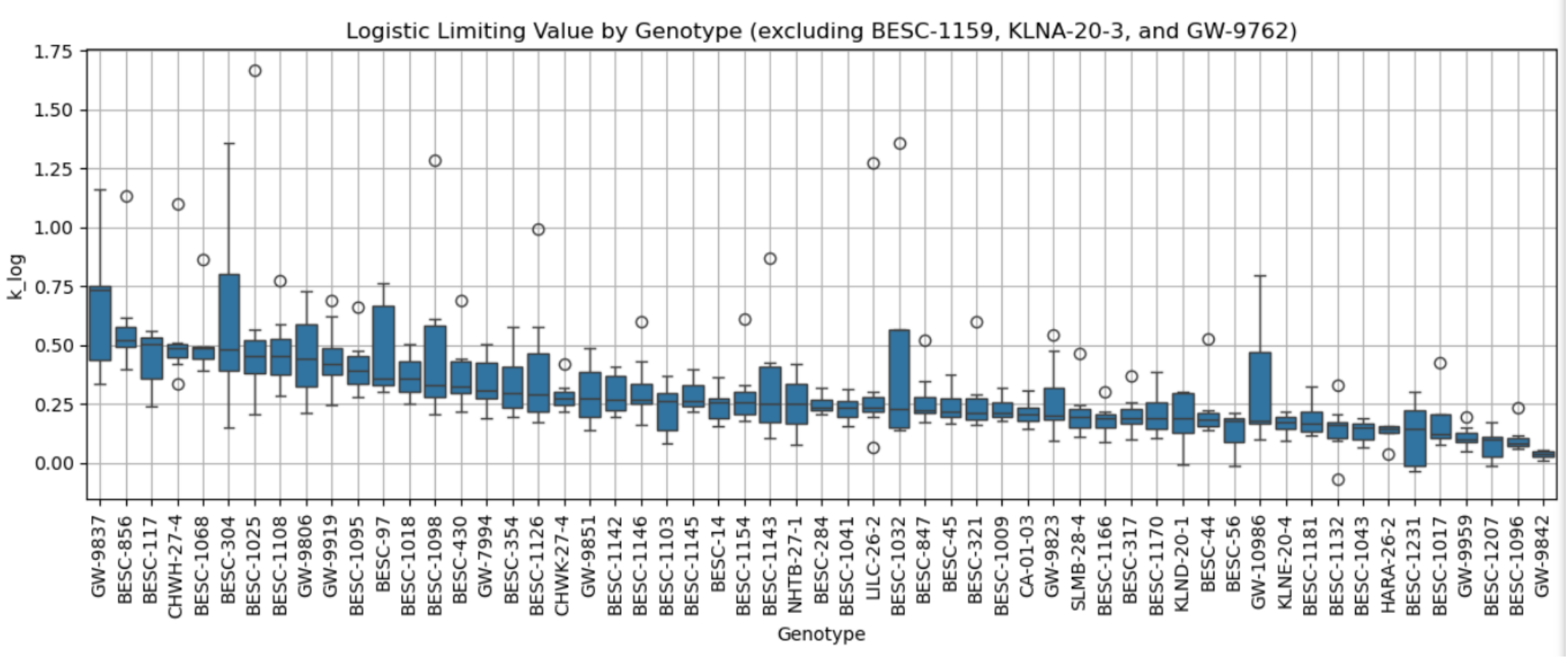
Takeaways:

Strong evidence that growth rate (r) differs by genotype.

Some genotypes consistently perform better or worse in early growth.

BESC-1132 consistently deviated from others.

ANOVA results for Logistic Limiting Value - Area



Takeaways:

Helps confirm that some genotypes grow fast not just because they start large.

ANOVA

```
===== ANOVA Table for r_log =====
                sum_sq     df         F    PR(>F)
C(genotype)  18.631933   59.0  1.457733  0.021338
Residual     77.555214  358.0       NaN       NaN

===== ANOVA Table for r_lin =====
                sum_sq     df         F        PR(>F)
C(genotype)  0.004185   59.0  7.575884  5.602642e-36
Residual     0.003352  358.0       NaN           NaN

===== ANOVA Table for k_log =====
                   sum_sq     df         F    PR(>F)
C(genotype)  9.907703e+06   59.0  0.849918  0.774459
Residual     7.073382e+07  358.0       NaN       NaN
```

→ There *is* a statistically significant effect of genotype on r_log at the 5% significance level.

→ Strong statistical evidence that genotype has a significant effect on r_lin.

→ No evidence that genotype influences k_log

Heritability Analysis

Trait	MS Genotype	MS Error	n_r	Var(G)	Var(E)	Var(P)	H^2
r_log	0.3158	0.2166	6.97	0.0142	0.2166	0.2309	0.062
r_lin	0.0001	0.0000	6.97	0.0000	0.0000	0.0000	0.486
k_log	167927.1713	197580.5123	6.97	-4256.4604	197580.5123	193324.0519	-0.022

Takeaways:
Heretibility is low for r_log and K_log, indicating that variation is not due to genotype.
Heretibility for r_lin is larger, indicating a good portion of variation is explained by genotype.

Section 4: Clustering Analysis

Methodology of Analysis

Weighted Growth Curves

- Using the best fitted model for each plant, calculate the weighted growth of the genotype.
- Weighted growth is calculated from the individual growth curves and weights curves farther from the median at time t less.

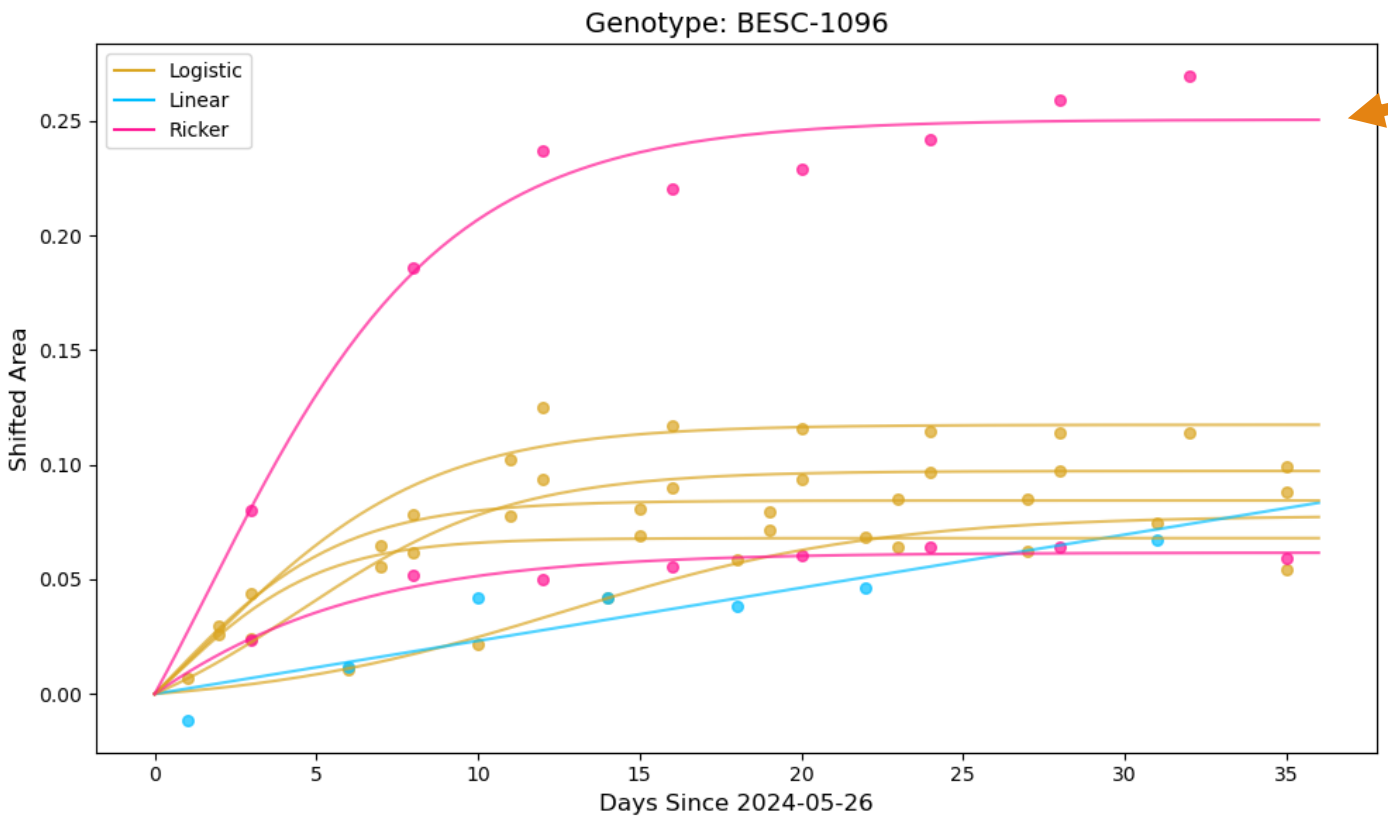
Weighted Fits

- Fit the five models to each weighted curve to extract growth parameters.
- Scale the resulting growth parameters using min-max scaling.

K-Means Clustering

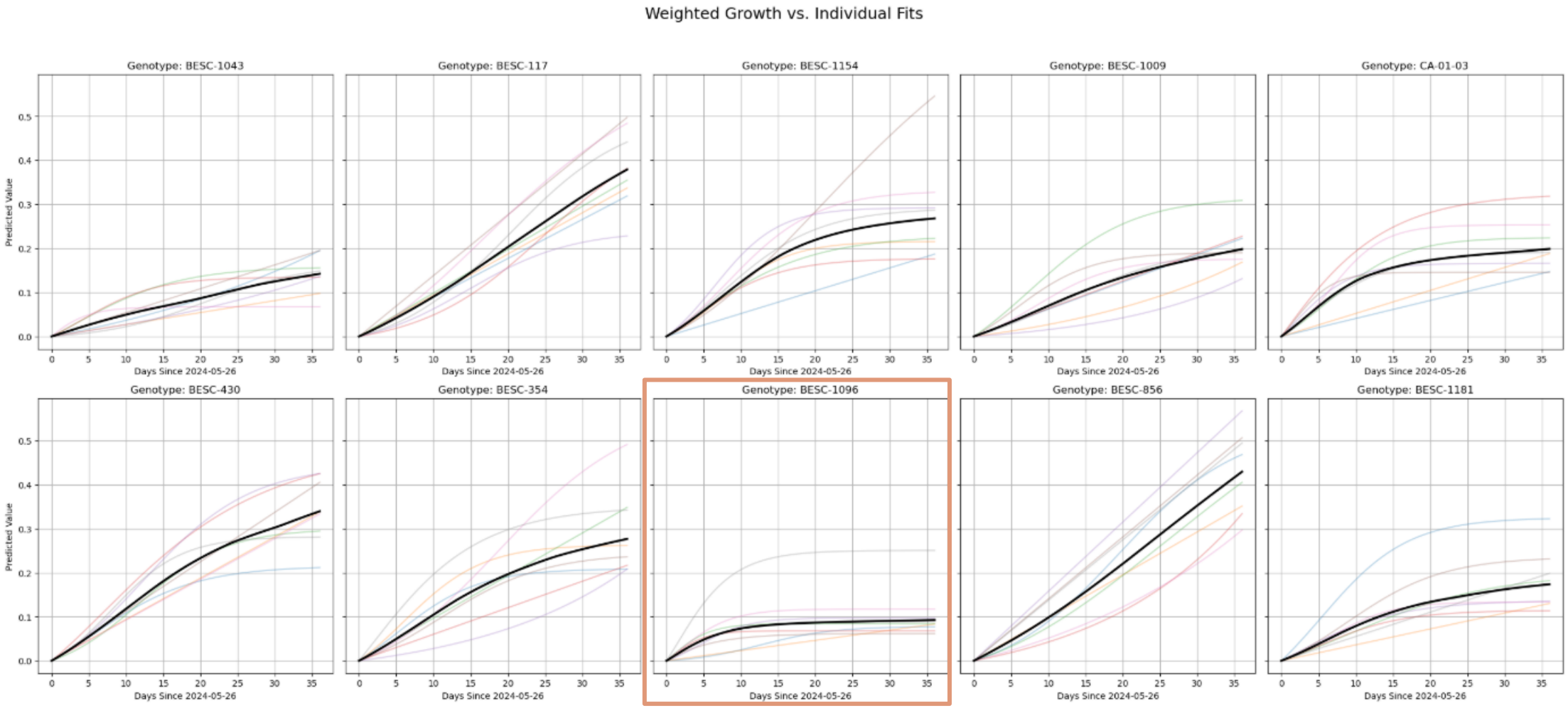
- Use K-Means++ initialization.
- Cluster using logistic function parameters *r* (growth rate) and *K* (limiting factor)

Analyzing "Average" Genotype Behavior



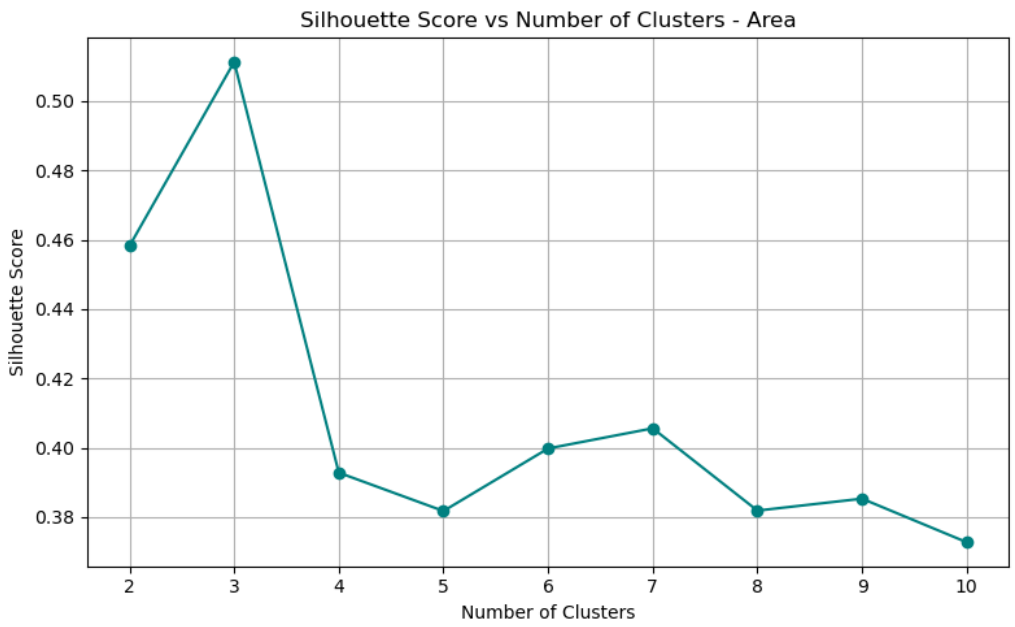
Outlier Plant within the BESC-1096 genotype

Example Plots of Weighted Curves – Area



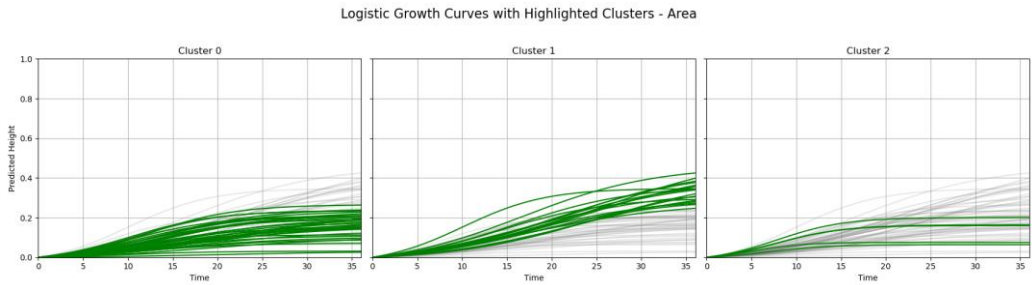
Clustering Analysis – Area

The optimal number of clusters for area was three with a silhouette coefficient of 0.51.



Takeaways:

- Reasonably Strong Clustering
- Identification of “middle”, “high”, and “low” growth genotypes



Summary

- Growth Model Fitting:
  - We fit several growth models (logistic, Ricker, exponential, Generalized Von Bertalanffy, linear) to min-max scaled plant height and area data. Logistic and linear models were most commonly selected as best fits across individual plants.
- Statistical Analysis:
  - ANOVA results informed us of major variation between genotypes.
  - Heritability informed us that the linear growth rate appears to have stronger genetic control, while logistic parameters may be more environmentally influenced.
- Clustering Analysis
  - Clustering informed us some genotypes consistently exhibited faster or larger growth patterns.

Potential Next Steps

- Clustering on All Data
- Further Investigate High Growth Genotypes
- Evaluate First and Last Time Points and Compare to Using Entire Time Series
- Incorporate Environmental Data (drought simulation, cold temperatures, low sunlight)
- Biomass Prediction